Prof. Patti Hamerski

Day 7

Revisiting Gen-AI Guidelines and Analyzing Fourier Frequencies

Spring 2025

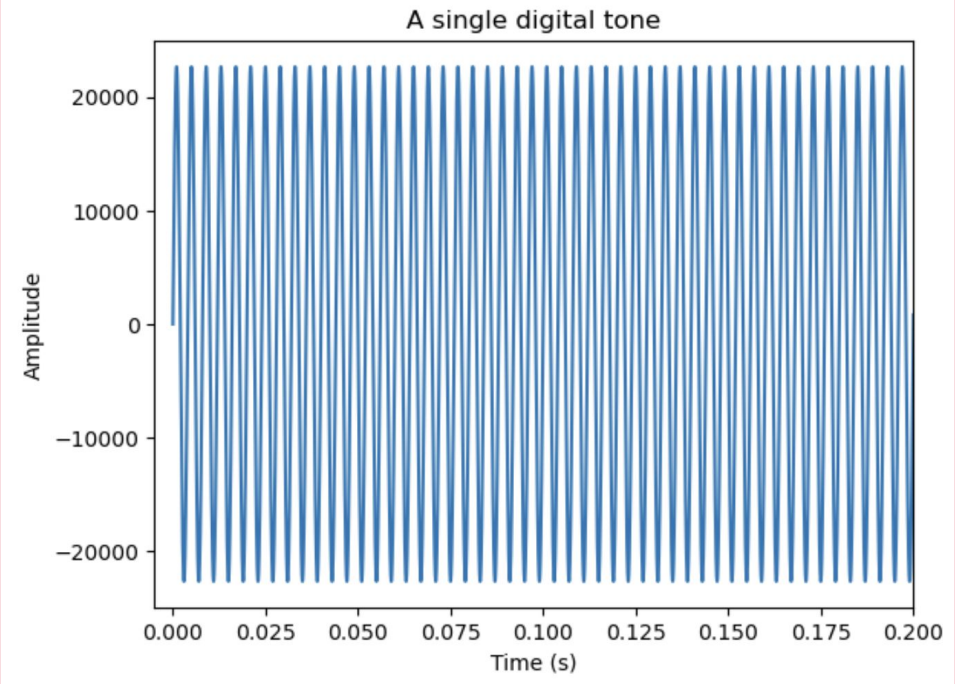
PH 367: Computational Physics Lab III

Today’s Class

- Short intro/teaser with Jeff on comparing musical notes
- Activity led by Pachi to begin finalizing class gen-AI guidelines
- In-class assignment comparing Fourier frequencies

Example: analyzing a single tone

A single 250 Hz tone:

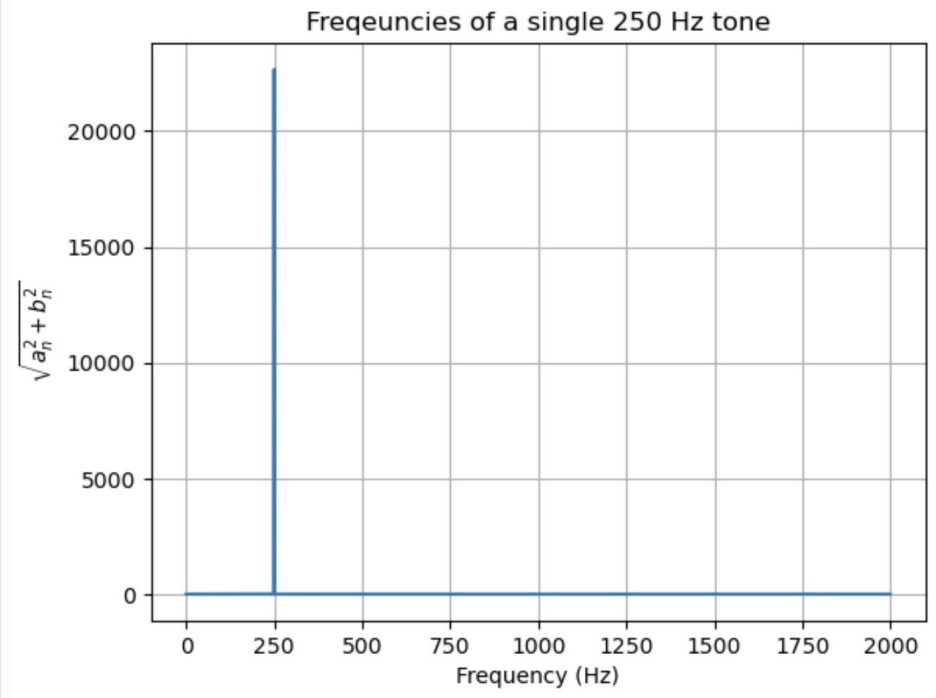


Calculating an and bn coefficients

$$a_n = \frac{2}{T}\int_0^T \cos\left(\frac{2\pi n t}{T}\right) f(t)\mathrm{d}t$$

$$b_n = \frac{2}{T}\int_0^T \sin\left(\frac{2\pi n t}{T}\right) f(t)\mathrm{d}t$$

```python
def anbn(f_vals, t_vals, n):
    T = t_vals[-1]
    trig_input = 2 * np.pi * n * t_vals / T
    an = 2 / T * integrate.trapezoid(np.cos(trig_input) * f_vals, x=t_vals)
    bn = 2 / T * integrate.trapezoid(np.sin(trig_input) * f_vals, x=t_vals)
    return an, bn
```

Frequencies of a single tone

Frequency distribution: